

Assignment 1: Fuzzy Sets and Relations: 20:08:2026

General Instructions

- Write a separate Python program for each question.
 - Matrix dimensions, membership values, learning rates, number of epochs, etc. should be accepted as input wherever applicable rather than being completely hard-coded.
 - For fuzzy sets and fuzzy relations, membership values must lie in the interval $[0, 1]$.
 - For crisp relations, membership values must be restricted to 0 or 1.
 - Display the input and the corresponding output clearly.
 - Do not use ready-made Fuzzy Logic or Neural Network libraries for implementing the core algorithms.
1. Implement the following basic operations on two fuzzy sets:
 - o Union
 - o Intersection
 - o Complement
 - o Difference
 2. Implement Fuzzy and Crisp Composition using both:
 - o Max-Min Composition
 - o Max-Product CompositionWrite four different programs/files:
 - o Crisp relation using Max-Min
 - o Crisp relation using Max-Product
 - o Fuzzy relation using Max-Min
 - o Fuzzy relation using Max-Product
 3. Write a program to check whether a fuzzy relation satisfies the fuzzy equivalence/similarity relation properties. The program must individually check:
 - o Reflexivity
 - o Symmetry
 - o Transitivity

If the relation is not an equivalence relation, the program must clearly display which property/properties are violated and the corresponding reason.

Allow the relation matrices and their dimensions to be entered by the user